

The Theory of Persistence: a reader's map

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Abstract

The Theory of Persistence (PT) derives the full phenomenology of the Standard Model from a single formal input, the arithmetic parameter $s = 1/2$. This preface gives the reader's map of the program: statements of the seven structural theorems **T0–T6**, of the six bridge axioms **BA0–BA5**, of the fixed point $\mu^* = 3 + 5 + 7 = 15$, of the prime taxonomy (active $\{3, 5, 7\}$, echoes $\{11, 13\}$, super-echoes $\{17, 19, 23\}$), of the arithmetic cascade principle, and of the canonical application to the fine-structure constant $1/\alpha_{\text{EM}} = 137,035\,999\,083$ at 0,004 ppb. No proof is given here; every statement points to the M1–M5 articles where the full derivation appears. This article serves as the pedagogical entry point of the series.

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1 Motivation: why a sieve?

The Standard Model of particle physics involves about twenty numerical parameters without theoretical derivation: three gauge couplings, the Higgs vacuum expectation value and its self-coupling, nine charged masses, three CKM mixing angles and one phase, plus the analogous neutrino-sector parameters [1]. These numbers are measured, not predicted.

The Theory of Persistence proposes that they all follow from a single arithmetic structure: the Eratosthenes sieve read as a dynamical system on the sequence $\{g_n\}_{n \geq 1}$ of consecutive prime gaps. The unique formal input of the theory is a rational number,

$$s = \frac{1}{2},$$

called the *persistence parameter*. It quantifies the symmetric fraction of information preserved by a system under successive constraints; its exact value $1/2$ coincides with the spectral gap of the primordial transfer matrix (theorem **T2**).

From this single number, PT derives seven structural theorems **T0–T6**, six proved bridge axioms **BA0–BA5**, an arithmetic fixed point $\mu^* = 15$, and the zero-parameter derivation of every accessible constant of the Standard Model. The program is laid out in five companion articles:

M1 [2] the sieve as a dynamical field: **T1, T3, T6**, lemma **L0**, Gap Fundamental Theorem.

M2 [3] information geometry: Fisher metric, holonomy **BA3**, activation criterion **BA4**.

M3 [4] convergence and fixed point: **T2, T4, T5**, closing **BA0** $\rightarrow$ **T0**.

M4 [5] extended structures: sieve algebra, complex mechanics, predictive mathematics.

M5 [6] bridge to physics: **BA5**, Pontryagin identification, reconstruction lemmas E/F/G, $\alpha_{EM} = 1/137,036$.

This preface gives the map. Two extension articles complete the program: [7] for the geometric-spectral readings (persistence curve, Ricci soliton), and [8] for the spectral-arithmetic RH-PT conjecture.

2 The PT sieve and its fixed point

2.1 Minimal formal framework

Let L^0 be the category of finite pointed sets over ZFC, equipped with cardinal additivity. The *maximum entropy lemma* **L0** ([2]) asserts that the unique distribution on $\mathbb{N}$ under a finite-mean constraint is the geometric distribution, parameterised by a real $q \in (0, 1)$. This distribution is precisely the asymptotic measure of the gaps $\{g_n\}$ between consecutive primes on the 6-rough sieve.

At the arithmetic fixed point μ^* (introduced in 2.3), the same geometric law admits *two equivalent canonical parametrisations*, related by the vertex–edge involution of the sieve graph:

$$q_+ = 1 - \frac{2}{\mu^*} = \frac{13}{15} \quad (\text{vertex branch, max-entropy}), \quad (1)$$

$$q_- = e^{-1/\mu^*} = e^{-1/15} \quad (\text{edge branch, Gibbs}). \quad (2)$$

The letter q is here the standard parameter of the geometric distribution in elementary statistics; the subscripts $\pm$ distinguish the two parametrisations.

2.2 The seven structural theorems **T0–T6**

Table 1 lists the seven theorems of the arithmetic core of PT, in the canonical order of the series' nomenclature map. All are [THM]: unconditionally proved.

2.3 The fixed point $\mu^* = 15$

Theorem **T5** proves that the cascade equation $\mu = \sum_{p \in \mathcal{A}(\mu)} p$, where $\mathcal{A}(\mu)$ is the set of primes p satisfying $\gamma_p(\mu) > 1/2$ (activation criterion, **BA4**), admits a unique positive integer solution:

$$\boxed{\mu^* = 3 + 5 + 7 = 15.}$$

At this value, $\mathcal{A} = \{3, 5, 7\}$ and the three anomalous dimensions are

$$\gamma_3 = 0,808, \quad \gamma_5 = 0,696, \quad \gamma_7 = 0,595.$$

The detailed proof and linear stability of the fixed point are in M3 [4].

Code	Name	Short statement	Article
T0	Dynamical Field Theorem	$\{g_n\}$ is the unique field consistent with conditions U_1 – U_4	M3
T1	Forbidden Transitions mod 3	$T_3[1, 1] = T_3[2, 2] = 0$ on 6-rough integers	M1
T2	Spectral Conservation	$ \lambda_2(T_{30}) = 1/4 = s^2$	M1
T3	Antidiagonal Transfer	$T_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, spectrum $\{1, -1\}$	M1
T4	Spectral Convergence	$\alpha_k \rightarrow 1/2$ as sieve depth $k \rightarrow \infty$	M3
T5	Unique Fixed Point	$\mu^* = 3 + 5 + 7 = 15$	M3
T6	Complete Uniqueness	Eratosthenes sieve unique (T6a + T6b + T6c)	M1

Table 1: The seven structural theorems of the PT arithmetic core.

3 Bridge axioms and prime taxonomy

3.1 The six bridge axioms BA0–BA5

Above the arithmetic core **T0–T6**, PT rests on six *bridge axioms* relating the sieve structure to physical phenomenology. Though historically named “axioms”, all are proved [THM]. Table 2 summarises.

Code	Name	Short statement	Article
BA0	Field axiom	Field = $\{g_n\}$, T0 promoted	M3, M5
BA1	Observable axiom	Observables = $g_n \bmod m$ (gauge connection)	M5
BA2	GFT (fundamental identity)	$\log_2 m = D_{\text{KL}}(P \ U_m) + H(P)$	M1
BA3	Holonomy formula	$\sin^2 \theta_p = \delta_p(2 - \delta_p)$, $\delta_p = (1 - q^p)/p$	M2
BA4	Active prime criterion	$\gamma_p(\mu) > s = 1/2$ selects $\mathcal{A}$	M2
BA5	Pontryagin product	$\alpha_{\text{sieve}} = \prod_{p \in \{3,5,7\}} \sin^2 \theta_p(q_+)$	M5

Table 2: The six bridge axioms of PT (all proved as theorems).

3.2 Prime taxonomy

The activation criterion **BA4**, applied at the fixed point $\mu^* = 15$, partitions the primes into four canonical classes (plus the parity channel $p = 2$).

Class	Primes	$\gamma_p(\mu^*)$ or role	Reference
Parity channel	$\{2\}$	antipodal involution D	BA2, T3
Active	$\{3, 5, 7\}$	$\gamma_p > 1/2$, cascade	BA4, T5
Echoes	$\{11, 13\}$	$\gamma \in (0, 1/2)$, universal vacuum-polarisation	M2, M3
Super-echoes	$\{17, 19, 23\}$	scale-dependent corrections (hadronic)	M3, M5
Inactive	$\{p \geq 29\}$	$F_{\text{inactive}}(\mu^*) \approx 10^{-4}$	M3

Table 3: Complete taxonomy of primes in PT, at the fixed point $\mu^* = 15$.

Strict canonical convention (cf. [3, 4]):

- Primes $\{11, 13\}$ are the *echoes*, and only the echoes. Never super-echoes.
- Primes $\{17, 19, 23\}$ are the *super-echoes*, never $\{11, 13\}$.

- For the sum of echo + super-echo contributions, the term *extended* β_{echo} is used.

4 The arithmetic cascade principle

4.1 Definition

The *canonical cascade* of PT is the ordered sequence of modular refinements on the three active primes:

$$\mathcal{C}_0 \xrightarrow{\gamma_3} \mathcal{C}_3 \xrightarrow{\gamma_5} \mathcal{C}_{3,5} \xrightarrow{\gamma_7} \mathcal{C}_{3,5,7}.$$

Each arrow weights the pattern by the corresponding anomalous dimension γ_p . The cumulative amplitude at stage j is $\Gamma_j = \prod_{i=1}^j \gamma_{p(i)}$; for the canonical cascade:

$$\Gamma_1 = 0,808, \quad \Gamma_2 = 0,562, \quad \Gamma_3 = 0,335.$$

The effective depth $\sum_{p \in \mathcal{A}} p = 3 + 5 + 7$ coincides with the fixed point $\mu^* = 15$; this is the geometric reading of theorem **T5**.

4.2 Local forcing

The two intra-PT bridge theorems BT₇ (Gram-isotropy of pure roles $\{\tau, f_3, f_5, f_7\}$) and BT₉ (local forcing by activation threshold) guarantee that the canonical cascade is *the unique* ordered sequence compatible with **BA4** and **T5**. Details are in M4 [5].

5 Canonical application: the fine-structure constant

The complete chain, with no adjustable parameter, leads from $s = 1/2$ to the numerical value of α_{EM} :

$$s = 1/2 \xrightarrow{\mathbf{T1-T6}} \mu^* = 15 \xrightarrow{\mathbf{BA3}} \sin^2 \theta_p(q_+) \xrightarrow{\mathbf{BA5}} \alpha_{\text{bare}} = \prod_{p \in \{3,5,7\}} \sin^2 \theta_p(q_+).$$

The numerical values (computed in M5 [6]) are

$$\sin^2 \theta_3(q_+) = 0,2192, \quad \sin^2 \theta_5(q_+) = 0,1940, \quad \sin^2 \theta_7(q_+) = 0,1726,$$

giving the *bare coupling*

$$\alpha_{\text{bare}} = 0,2192 \times 0,1940 \times 0,1726 = \frac{1}{136,28}.$$

The correction by the parity channel ($p = 2$) — binary leakage, spiral retroaction, screening by echoes $\{11, 13\}$, two-loop vacuum polarisation — closes the gap to the measured value. The final result, with no adjustable parameter [6]:

$$\frac{1}{\alpha_{\text{EM}}^{\text{PT}}} = 137,035\,999\,083 \quad (\text{deviation: } 0,004 \text{ ppb}).$$

The measured value from the quantum Hall effect and electron anomalous magnetic moment [1] is

$$\frac{1}{\alpha_{\text{EM}}^{\text{exp}}} = 137,035\,999\,084(21),$$

agreement to experimental precision (0,15 ppb).

6 Reading plan

Level 1 (reference arithmetic corpus): the series M1→M5 develops the complete theory with proofs. It is designed for sequential reading.

- M1 [2]: dynamical sieve, **T1**, **T3**, **T6**, **L0**, GFT.
- M2 [3]: information geometry, holonomy **BA3**, active primes.
- M3 [4]: convergence **T4**, fixed point **T5**, closing **T0**.
- M4 [5]: extended structures, predictive mathematics.
- M5 [6]: arithmetic-to-physics bridge, **BA5**, reconstruction lemmas $E/F/G$, α_{EM} .

Level 2 (thematic extensions):

- [7]: geometric and spectral readings — algebraic persistence curve of genus $\mu^* - 1 = 14$, Fisher–Bianchi metric as Ricci quasi-soliton, spectral identities K2 and A6.
- [8]: spectral-arithmetic RH-PT conjecture, residual survival operator.

Epistemic status. The seven theorems **T0–T6** and the six bridge axioms **BA0–BA5** are all proved ([THM]). The PT program closed in April 2026 its ten critical structural commitments (counter *C1–C10*, 10/10 [THM] after the C2/C4/C5/C7 promotions of 26 April 2026). The only remaining conditional step at the program level is external peer review; this article series constitutes its input material.

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